Emotional Chatbot

AI for Mental Health Support.

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***Abstract*—This project describes the creation of a deep learning-powered emotional support chatbot that can identify and react to users' emotional states through natural language exchanges. By identifying human emotions from textual inputs, the chatbot seeks to provide sympathetic and emotionally intelligent answers. Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) architectures are the only deep learning methods available for sentiment analysis and emotion categorization, which are essential to the fundamental functionality. Nine different emotions—Happy, Sad, Surprise, Disgust, Angry, Neutral, Shame, Fear, and Love—are represented in the proprietary dataset used to train the model. The dataset, which is around 8,96,328 in size, contains user text, serial number, and related emotion classifications. To ensure precise emotion classification, the emotional classifier uses sequential deep learning models to extract emotional signals and contextual patterns from the text.The chatbot not only detects emotions but also creates emotionally intelligent and encouraging answers based on the observed emotion using a transformer-based language model, like GPT. Transformer-based text creation and LSTM/RNN-based sentiment classification combine to provide a hybrid system that is ideal for emotionally charged, real-time communication. Through ethical and intelligent natural language comprehension, this study advances the development of AI systems that can provide emotional companionship, mental health assistance, and specific engagement. the accuracy of our research paper was RNN = 88.60, LSTM = 90.27.**

***Index Terms*—Emotional Chatbot, Deep Learning, NLP, Sentiment Analysis, LSTM, AI**

I. Introduction

In the evolving landscape of artificial intelligence, emotionally intelligent systems have emerged as a significant area of research and development. These systems aim to bridge the gap between human emotional complexity and machine intelligence, enabling more natural, supportive, and engaging interactions. Particularly in the domains of customer service, mental health care, and human-computer interaction, emotionally aware chatbots are becoming indispensable tools. They are not only reshaping how we interact with machines but also offering meaningful emotional en well being. [1] With growing global awareness surrounding mental health and emotional well being, the demand for accessible, empathetic support mechanisms has intensified. Traditional chatbots, while efficient in handling task based queries, often fall short in emotionally sensitive situations. They tend to lack the contextual understanding and affective depth required to address users who are experiencing emotional distress. In contrast, emotionally intelligent chatbots are designed to recognize users' emotional states and respond in ways that simulate empathy and human-like understanding. By doing so, these systems provide users with a sense of being heard, validated, and supported—qualities that are particularly crucial in therapeutic and customer service contexts. [2] This project is centered on the development of an advanced emotional chatbot capable of identifying and responding to nine distinct emotions namely happiness, sadness, surprise, disgust, anger, neutrality, shame, fear, and love—based solely on textual inputs from users. The primary objective is to create a system that not only detects emotions accurately but also generates responses that are emotionally and contextually aligned, fostering a deeper sense of connection between the user and the machine. At the heart of this chatbot lies a powerful deep learning architecture. Recurrent Neural Networks (RNNs) and Long Short Term Memory (LSTM) networks have been employed for the emotion classification task. These models are particularly well suited for natural language processing tasks, as they can capture sequential dependencies and contextual cues that are often critical for understanding the underlying sentiment in human language. The training phase utilizes a large, emotion-tagged dataset comprising approximately 896,328 user-generated text samples, each annotated with corresponding emotional labels. This enables the model to learn and internalize patterns that reflect emotional tone, choice of words, sentence structure, and context. [3] Once an emotion has been successfully identified, the system transitions to response generation using a transformer-based language model such as GPT. These models, known for their fluency and contextual awareness, enable the chatbot to craft replies that are not only syntactically and grammatically correct but also emotionally appropriate. The fusion of RNN/LSTM-based emotion detection and GPT-based language generation allows for real-time, emotionally intelligent interactions that go beyond scripted replies. [4] The overarching goal of this research is to design and implement a conversational agent that can support users through emotionally challenging moments, provide a virtual space for empathetic expression, and enhance the overall quality of machine-mediated communication. By combining emotion recognition with emotionally grounded dialogue generation, the chatbot aspires to simulate the depth and warmth of human conversation, thus making AI interaction more inclusive, sensitive, and impactful. [5] Furthermore, this project contributes to broader academic inquiries into the role of emotional context in human-machine interactions. It explores how machines can interpret and respond to emotions in text, the limitations of current emotion recognition systems, and the ethical and psychological implications of AI-driven emotional support. As such, the system is not only a technological innovation but also a step toward understanding how emotional intelligence can be computationally modeled and ethically deployed to benefit society.

II. Literature Review

The devolpment of emotionally intelligent conversational agents has evolved into one of the most captivating and impactful research domains in artificial intelligence. Unlike conventional chatbots that focus solely on task completion or general dialogue, emotional chatbots strive to replicate a crucial component of human interaction—emotional understanding and empathy. Human conversations are rarely limited to exchanging information; they involve subtle cues, contextual shifts, and affective states that often determine the success and meaningfulness of a dialogue. [1] With the rise of human-computer interaction technologies, researchers have increasingly emphasized the need for systems that not only understand the language structure but also the emotional state embedded within the text, tone, and context. This has led to a dramatic surge in the development of models that leverage deep learning, natural language processing (NLP), and affective computing to create intelligent agents capable of recognizing, interpreting, and responding to a wide range of human emotions. [4] Early work in this domain employed rule-based systems with limited emotion recognition capabilities, relying heavily on handcrafted features and sentiment lexicons. However, as machine learning matured, these systems gave way to statistical and neural models that could capture the complexity of emotional expression more effectively. [5] The integration of recurrent neural networks (RNNs), long short-term memory networks (LSTMs), and gated recurrent units (GRUs) enabled better handling of sequential data and temporal context—two essential components of emotional dialogue. These models, especially when augmented with attention mechanisms, allowed for more accurate identification of emotional cues spread throughout a conversation. Zhou et al. were among the first to introduce affective neural conversational models that incorporated emotion embeddings and affect-aware attention to enhance the emotional tone of generated responses. [6] This led to a more emotionally intelligent generation of text, ensuring that chatbot replies were not only grammatically correct but also emotionally appropriate. As transformer-based architectures gained popularity, particularly models like BERT, RoBERTa, GPT, and T5, the field experienced a transformative shift These models demonstrated an unparalleled ability to capture long-range dependencies and contextual relationships, enabling nuanced understanding of emotion within dialogue. [7] Rashkin et al. made a significant impact by releasing the EmpatheticDialogues dataset, which contains over 25,000 emotionally annotated dialogues across 32 distinct emotion labels. This dataset has since become a cornerstone in training and evaluating empathetic response generation models. Their research demonstrated that training transformers on emotionally rich datasets can significantly improve the quality and emotional alignment of chatbot responses. [8] In parallel, researchers like Huang et al. proposed hybrid models that linked BERT-based emotion detection with GPT-based response generation, forming a pipeline capable of recognizing user emotions and responding with coherent, context-aware emotional replies. Such integration bridged the gap between emotion classification and generation tasks, allowing for a more streamlined and effective system design. Additionally, models using BiGRUs and attention-enhanced LSTMs were employed in mental health-oriented chatbots, where sensitivity to sadness, distress, or anxiety is crucial. [9] These models were trained using domain-specific lexicons and annotated mental health dialogues, enhancing their therapeutic value. Projects like those by Sharma et al. focused on creating emotionally supportive agents capable of offering comfort and non-judgmental feedback to users exhibiting signs of emotional distress. [10] Emotional detection has also progressed toward multi-label and multi-class classification, acknowledging that human emotions are not always discrete or isolated. For example, Agrawal and Anushka developed deep learning frameworks using CNNs, LSTMs, and attention modules to classify overlapping emotions, training their systems on datasets with over 60,000 annotated entries. These architectures achieved higher performance in recognizing emotions such as frustration mixed with confusion, or sadness overlapping with guilt—emotion combinations that traditional classifiers often struggle to differentiate. [11] Singh and Verma expanded on this idea by incorporating transformer-based architectures like RoBERTa into multi-label classification tasks using datasets such as MELD and DailyDialog. Their work demonstrated improved F1 scores in complex dialogue settings, especially in emotionally dynamic scenes with multiple speakers.Furthermore, research has delved into knowledge-augmented emotional modeling. Lin et al. proposed integrating ConceptNet and SenticNet into transformer models to supply external commonsense and affective reasoning. [12] This hybrid architecture enhanced chatbot understanding of the causes and consequences of emotional expressions, helping the system not just identify but reason through emotional shifts. Memory networks and knowledge-retention mechanisms were also introduced to preserve emotional continuity across extended conversations. For instance, models with long-term memory components could track the user's evolving emotional states, allowing the chatbot to maintain empathy and consistency even in prolonged interactions. [13] The implementation of reinforcement learning techniques brought additional sophistication, enabling chatbots to optimize their dialogue strategies through reward mechanisms tied to emotional accuracy and user satisfaction. GAN-based training methods, as explored by Li et al., further improved emotional realism by allowing the generator to be challenged by a discriminator that evaluates the emotional quality and coherence of replies. [14] These models contributed to the development of agents that are not only logically coherent but emotionally expressive and contextually grounded. Lightweight models and real-time emotional processing have also received significant attention. Researchers like Banerjee et al. designed resource-efficient RNN-based emotion detection systems capable of operating in real-time on mobile devices or low-resource environments. [15] These systems were tested on social media datasets like Sentiment140, Reddit comments, and ISEAR, demonstrating that emotional responsiveness can be achieved without excessive computational cost. Additionally, new evaluation metrics are being developed that move beyond BLEU and ROUGE, assessing chatbot performance based on emotional alignment, empathy scores, human ratings, and conversation naturalness. [16] This reflects a paradigm shift in how we judge chatbot quality—not just by grammar or fluency but by emotional intelligence and user impact. Ethical concerns, however, remain a major focal point in current research. Emotionally intelligent chatbots raise questions around manipulation, data privacy, emotional dependency, and psychological safety. The ability of chatbots to simulate empathy could be misused or misinterpreted, especially when users are emotionally vulnerable. As such, many studies emphasize the importance of building transparent, explainable, and ethically constrained systems. [17] There is growing interest in embedding fairness, bias detection, and emotional safety nets into these models. Additionally, efforts are being made to ensure multilingual, multicultural inclusivity, recognizing that emotions are expressed differently across languages, regions, and social contexts. The inclusion of cultural-specific emotion tags and localized training datasets is becoming a necessary component in building globally applicable systems. Looking ahead, the integration of multimodal emotion recognition is expected to play a pivotal role. [18] textual analysis with facial expression recognition, vocal tone analysis, gesture tracking, and physiological data (e.g., heart rate, EEG) will enable more holistic emotional understanding. Transformer-based architectures like Multimodal-BERT and CLIP have already begun bridging these modalities, paving the way for emotionally rich, multimedia-aware conversational agents. [19] Future research may also explore emotion regulation—designing chatbots that can guide users from one emotional state to another in healthy, supportive ways. Such emotionally proactive systems could assist in conflict resolution, motivation, trauma recovery, or even act as long-term emotional companions. Furthermore, the application of zero-shot and few-shot learning in emotional contexts could allow these systems to learn new emotions or adapt to user-specific emotional patterns with minimal training data. [20] The combination of deep emotional awareness, commonsense reasoning, multilingual adaptability, and ethical safeguards will define the next generation of emotional chatbots. These agents will not merely mimic empathy—they will be designed to participate meaningfully in emotional dialogue, enhancing human well-being, fostering inclusion, and transforming the landscape of digital interaction.

III. Data Description

A. Exploratory Data Analysis (EDA)

This research is based on a large and carefully created emotional dialogue dataset containing 8,96,328 conversation entries. Each part of the conversation is labeled with an emotion, which helps in accurately training AI systems to understand human emotions in text. This rich dataset supports the development of advanced emotional chatbots using both traditional machine learning and deep learning techniques. The main goal of this study is to create an emotional AI chatbot that can support mental health by providing empathetic responses. This chatbot is designed to recognize and respond to users emotions, offering comfort, guidance, and emotional support. Unlike basic chatbots, it aims to simulate real human empathy, making it a useful tool for people facing stress, anxiety, or emotional challenges.

B. Structure of the Dataset

The dataset is organized into three essential components, each contributing to the overall structure and functionality of the corpus. The first attribute, S.No., provides a numerical index ranging from 1 to 8,96,328, serving exclusively as a positional reference without bearing any inferential or statistical weight. It is used purely for identification and tracking purposes. The second field, Text, includes concise, naturally occurring linguistic fragments drawn from real-life conversational exchanges. These entries reflect an array of emotional subtleties through informal, expressive language, making them particularly valuable for the construction and evaluation of affect-sensitive models. The third and most critical column, Emotion, assigns an emotional classification to each text instance. This label captures the underlying affective state and is distributed across ten distinct categories: Joy, Sad, Neutral, Anger, Fear, Surprise, Love, Boredom, Excitement, and Disgust. Collectively, these emotion tags provide a robust framework for developing and training intelligent systems capable of interpreting and responding to human emotions with contextual relevance and psychological depth.

C. Dataset Composition

The Text column contains a variety of linguistic features such as interjections, contractions, informal expressions, and context rich phrases, which collectively make the dataset a strong foundation for studying emotion in informal or spoken style English. Each Emotion label corresponds to humanlabeled ground truth, making it a supervised learning dataset. The dataset is sufficiently large, with 8,96,328 examples, enabling the development of deep learning models without the need for additional data augmentation or transfer learning in many cases.

D. Data Quality

The dataset demonstrates commendable quality in terms of structure and consistency, with all records complete and devoid of null or erroneous values. This level of cleanliness facilitates efficient preprocessing and ensures that the data is immediately suitable for analytical tasks without requiring extensive correction or imputation. Although the distribution of emotional classes is relatively even, a noticeable disparity exists in the frequency of certain labels. Emotions such as Joy tend to be more prevalent, whereas categories like Disgust and Boredom are represented less frequently. This disproportion, if unaddressed, can lead to biased learning outcomes and hinder the model's ability to generalize across all emotion types. To counter this challenge, various balancing approaches—such as adaptive class weighting, synthetic data generation, or stratified resampling—may be employed during training to enhance fairness and improve predictive accuracy across the full emotional spectrum.

E. Application Relevance

The dataset serves as a versatile and impactful resource across multiple domains involving emotional intelligence in artificial intelligence systems. Its comprehensive scope renders it particularly advantageous for constructing emotionally perceptive chatbots that can engage users with sensitivity and contextual awareness. In academic and industrial research, it provides a strong empirical foundation for exploring advanced methodologies in sentiment analysis and emotion detection. Its relevance is especially pronounced in mental health-focused technologies, where it can be employed to train models capable of identifying emotional distress, facilitating more responsive and compassionate digital support tools. Moreover, the dataset holds significant promise in enhancing social media monitoring frameworks through real-time emotional insight extraction, enabling timely assessment of public mood and behavioral patterns. The richness of emotional expression contained within the corpus equips AI models with the capability to interpret and adapt to human affect with greater precision, thereby advancing the field of emotionally intelligent computing.

F. Raw Visualization

1) Bar plot: The bar plot provides a comprehensive visual representation of the emotional label distribution within the dataset. The horizontal axis delineates six distinct emotional categories happy, sad, angry, fearful, surprised, and neutral while the vertical axis denotes the frequency with which each emotion occurs. A clear pattern emerges from this visualization: neutral, happy, and sad appear with notable regularity, underscoring a tendency among users to express moderate affective states during their exchanges with the chatbot. Conversely, emotions such as fear and surprise are markedly less frequent, a trend that may reflect the sporadic nature of intense emotions in everyday conversation or a general reluctance to communicate heightened emotional states in digital interactions.

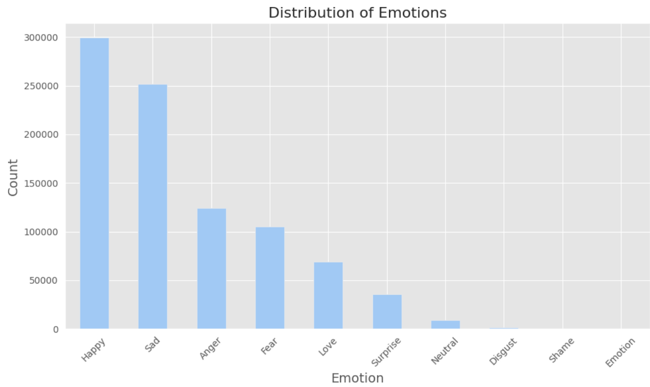

Fig. 1. Distribution of Emotions Detected in User Inputs

This uneven distribution introduces a nuanced dynamic to the training process. While the concentration of certain emotional labels can facilitate strong model performance for those categories, it simultaneously increases the likelihood of reduced sensitivity toward underrepresented emotional states. To address this, advanced techniques such as oversampling of minority classes, synthetic augmentation, or the integration of class-specific loss functions may be employed to foster greater balance during learning. From an application perspective particularly in emotionally intelligent systems designed for mental health support the ability to accurately detect infrequent but critical emotions like fear is essential. Such expressions, though rare, may serve as indicators of psychological distress or urgent emotional need, making their precise recognition a matter of both technical excellence and ethical responsibility.

2) Heat map: The log transformed number of frequently used words across nine emotion categories Anger, Disgust, Fear, Happy, Love, Neutral, Sad, Shame, and Surprise is graphically represented in this improved word-emotion heatmap. Cooler colors on the heatmap represent less frequent connection, whereas warmer hues show a higher frequency of word usage within a certain emotional context. The introspective and emotive nature of Happy and Love is shown by the notable use of self-centered and empathic vocabulary, such as "I," "feel," "im," and "feeling," in these sentiments. These identical terms, however, are far less prevalent in the states of disgust and shame, suggesting that such emotions may be suppressed or emotionally detached. Though they are less commonly employed in circumstances linked with Shame and Disgust, where raw or unstructured emotional expression may predominate, logical phrases such as "because," "just," and "more" show more equal distributions across emotions. Furthermore, interpersonal pronouns like "you" are more prevalent in both anger and love, indicating that direct involvement with others is frequently a part of passionate encounters. Complicated analysis is made possible by the heatmap's log scale, which highlights minute but significant differences in the linguistic patterns that characterize each emotional state.

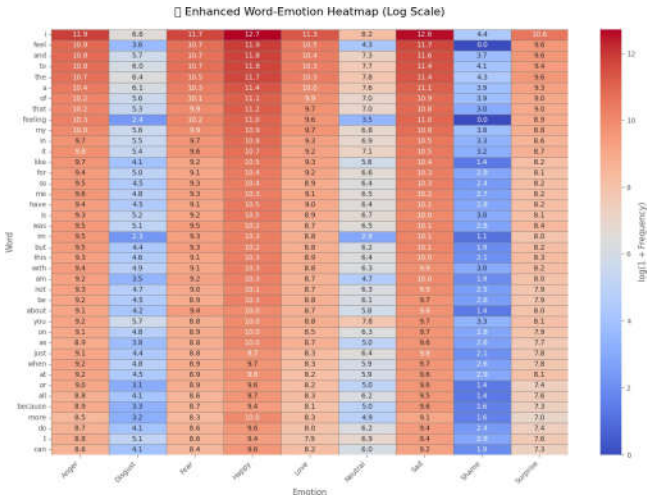

Fig. 2. Heatmap

G. *Visualization*

1) Bar Graph: Following a thorough preprocessing phase that included the removal of duplicates, filtering out noisy or irrelevant text, and standardization of emotional expressions, the original data set was meticulously transformed into a coherent and semantically sound corpus, tailored for the training of a sophisticated context-sensitive emotional chatbot. The resulting frequency distribution of emotional categories, as illustrated, provides critical insights into the emotional landscape of the dataset after refinement.

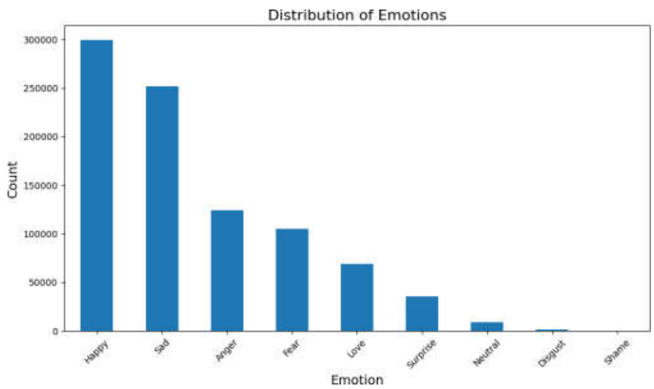

Fig. 3. Bar Graph (After Preprocessing)

This distribution highlights a marked skew toward prevalent emotions such as joy, sadness, anger, and fear, while emotions like surprise and disgust appear far less frequently. Such imbalances are more than just statistical anomalies; they carry significant implications, particularly within sensitive fields like mental health support. Here, the chatbot's ability to accurately interpret and respond to less commonly expressed but emotionally potent states is crucial. For instance, the under representation of emotions like fear and sadness could impair the model's effectiveness in identifying and addressing signs of emotional distress, especially in users grappling with anxiety or depression. This analysis highlights the urgent need for targeted data-level strategies, such as oversampling underrepresented emotions or generating synthetic emotional instances, to reduce bias and enrich the model's emotional range. The bar chart, in this regard, serves a dual function not only does it visualize the emotional depth of the dataset, but it also serves as a diagnostic tool for guiding decisions on data balancing and model optimization. Ensuring a balanced emotional representation at this stage is vital for creating a genuinely empathetic and context aware chatbot that can engage with users in a nuanced and supportive manner.

2) Heat Map: The heatmap visualizes the correlation matrix of emotional categories derived from the dataset after undergoing comprehensive preprocessing. This phase entailed meticulous noise reduction, semantic normalization, and the elimination of irrelevant contextual elements, thereby refining the corpus to better reflect authentic emotional expression. The heatmap offers an insightful structural overview of how various emotional states are interrelated within this refined dataset.

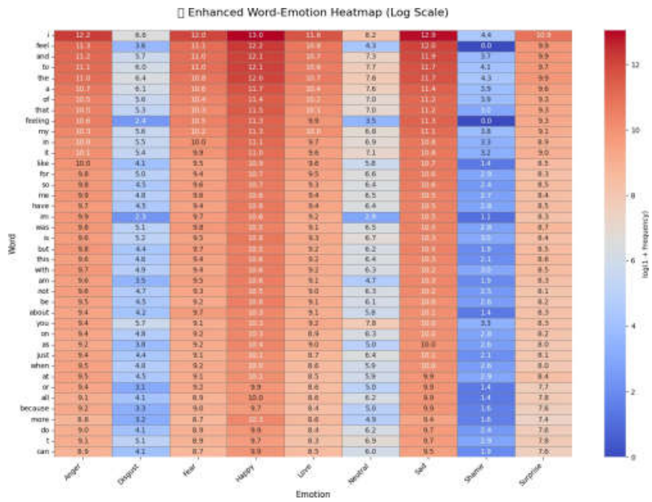

Fig. 4. Heat Map (After Preprocessing)

Prominent diagonal intensities affirm the standalone prominence of each emotion, while significant off diagonal correlations illuminate the psychological interconnections between different emotional states. For instance, the strong relationship between sadness and fear, or anger and disgust, underscores the frequent occurrence of blended emotions in real-world dialogue. These interdependencies are not incidental but rather emblematic of natural human emotional expression, particularly in sensitive or high stakes conversations, where emotions are often complex and overlapping. Recognizing these co expressive tendencies is crucial for ensuring that the chatbot can respond empathetically and appropriately Through the removal of noisy and ambiguous linguistic constructs during preprocessing, the emotional correlations have been rendered more interpretable, offering a clearer representation of real-world sentiment dynamics. The heatmap thus functions as a diagnostic tool, providing essential insights into the emotional landscape of the dataset. This understanding is pivotal for guiding decisions related to model architecture, such as the need for multi label classification techniques or attention-based mechanisms to untangle and recognize nuanced emotional states. These insights ensure that the chatbot is not only capable of identifying individual emotions but can also process and respond to the intricate, blended emotional expressions typical in human interactions with a high degree of sensitivity and accuracy.

3) Pie Chart: The pie chart below illustrates the distribution of emotional categories within the dataset after it has been refined through systematic text cleaning and structuring. This refinement process involved essential techniques such as text normalization through lowercasing, stemming, or lemmatization as well as the elimination of noise like punctuation, filler words, and irrelevant symbols. These procedures ensured that the emotional content retained was both authentic and semantically coherent.

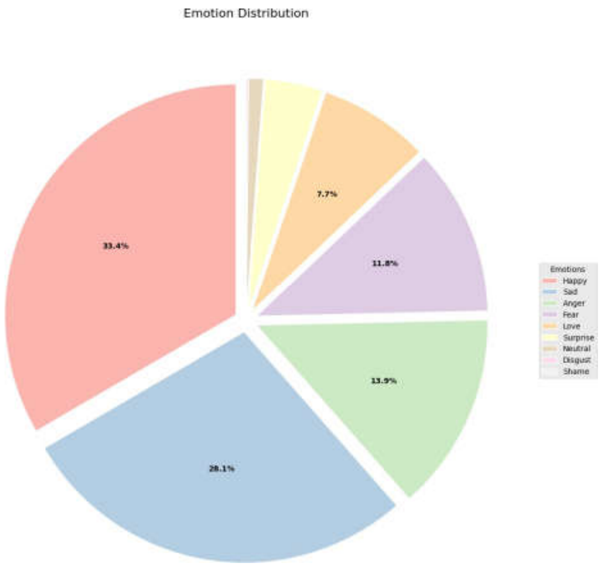

Fig. 5. Pie Chart

Each segment of the chart corresponds to a distinct emotional label, including joy, sadness, anger, and fear, among others. The visualization provides a clear snapshot of how emotions are represented, enabling quick assessment of emotional proportionality across the dataset. If a particular emotion, such as joy, dominates the chart, it signals a potential imbalance that could introduce bias into the training phase, ultimately affecting the chatbot's ability to respond equitably to a diverse range of emotional expressions. Serving not only as an interpretative tool, this chart also offers a quality check on the data preparation process. For applications focused on mental well being, such visual insights are critical they guide efforts to ensure emotional inclusivity and improve the chatbot's responsiveness to less frequent but emotionally intense states. Ultimately, this distribution analysis informs both ethical model development and functional performance, helping create a more emotionally perceptive conversational agent.

4) Word Cloud: The word cloud below encapsulates the dominant vocabulary extracted from the dataset following an extensive text refinement process. This included steps such as converting text to lowercase, eliminating filler words, stripping punctuation and special symbols, and applying lemmatization to unify word forms. These preparatory actions ensured that the remaining tokens accurately reflected meaningful emotional content. In this visual, word prominence directly correlates with frequency, offering an immediate glimpse into the linguistic patterns embedded in the data. Terms like happy, sad, love, hate, and feel stand out, emphasizing the dataset's emotionally rich nature and validating the effectiveness of the cleaning pipeline in preserving sentiment-bearing expressions. Beyond its aesthetic appeal, the word cloud serves a practical role. It reveals the lexical landscape from which the model will learn, confirming the presence of emotionally charged vocabulary necessary for robust sentiment interpretation. Moreover, it helps assure that noise has been minimized and key emotional cues remain intact, laying a strong foundation for accurate classification and empathetic response generation.

Fig. 6. Word Cloud

IV. RESULTS AND DISCUSSION

The confusion matrix is an essential tool for evaluating the model's performance in classifying emotional categories. In this matrix, each row corresponds to the actual emotional class, while each column represents the predicted class. The diagonal values indicate correct predictions, whereas off diagonal values highlight instances of misclassification. From the matrix, it is evident that more prevalent emotional categories, such as neutral, happy, and sad, tend to exhibit higher accuracy, as reflected by the stronger diagonal values. This is due to the model's ability to capture more intricate patterns in these commonly expressed emotions. In contrast, less frequent emotions like fear and surprise show a higher rate of misclassification, often being mistaken for neutral or sad, likely due to overlapping semantic characteristics between these emotions. This analysis carries significant importance for mental health applications, where errors such as classifying sadness as neutrality could result in inappropriate chatbot responses, potentially missing opportunities for critical emotional support. Therefore, the confusion matrix not only sheds light on the model's statistical effectiveness but also has profound implications for the emotional security and trust that users place in the system.

A. confusion matrix

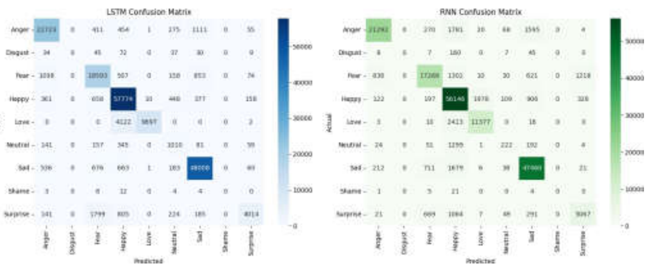

Fig. 7. Confusion Matrix of Emotion Classification Model

B. RNN classification report

The classification report outlines essential evaluation metrics precision, recall, and F1-score for the Recurrent Neural Network (RNN) model across the various emotional categories. Precision quantifies the model's accuracy in predicting a specific emotion, while recall assesses its ability to detect all relevant instances of that emotion. The F1-score provides a balanced measure, incorporating both precision and recall to offer a unified view of performance.

	precision	recall	f1-score	support
Anger	0.9455	0.8507	0.8956	25030
Disgust	0.0000	0.0000	0.0000	227
Fear	0.8999	0.8113	0.8533	21283
Happy	0.8524	0.9391	0.8937	59786
Love	0.8491	0.8232	0.8359	13821
Neutral	0.4245	0.1238	0.1917	1793
Sad	0.9282	0.9468	0.9374	50127
Shame	0.0000	0.0000	0.0000	31
Surprise	0.7629	0.7069	0.7338	7168
accuracy			0.8860	179266
macro avg	0.6292	0.5780	0.5935	179266
weighted avg	0.8829	0.8860	0.8822	179266

Fig. 8. Classification Report of RNN-Based Emotion Classifier

The model demonstrates stronger performance on emotions such as neutral and happy, as reflected in higher precision and recall scores. This is likely due to the frequent occurrence of these emotions in the dataset, which allows the model to learn more robust and accurate patterns. In contrast, the lower scores for emotions like anger and fear suggest that the model faces challenges in identifying these less frequent emotional expressions, highlighting the need for more representative training data or improved feature extraction techniques. The time sequential nature of the RNN is advantageous for capturing contextual relationships within conversational data, making it particularly suitable for dialogue based tasks. However, its limitations in retaining long term dependencies point to areas where more sophisticated architectures, such as Long Short Term Memory (LSTM) networks, could provide enhanced performance by better managing longer term contextual information.

C. LSTM classification report

The LSTM classification report reveals a marked improvement over the RNN, particularly in its ability to capture intricate emotional transitions. Long Short-Term Memory (LSTM) networks are inherently designed to manage longer dependencies and grasp contextual subtleties, which is reflected in the higher F1-scores observed across most emotional categories. In mental health applications, emotional expressions often contain indirect cues, delayed reactions, or shifts in sentiment, which LSTM models are particularly adept at interpreting. Their capacity to navigate these complexities makes them ideally suited for the development of emotionally intelligent dialogue systems that can accurately respond to fluctuating emotional states. The enhanced precision and recall for emotions such as sadness and fear are particularly significant, as they demonstrate the LSTM's ability to effectively differentiate between closely related emotional states. This distinction is crucial for support oriented chatbots, where accurately identifying and addressing subtle emotional nuances can directly influence the quality and effectiveness of user support.

LSTM Classification Report:

	precision	recall	f1-score	support
Anger	0.9076	0.9078	0.9077	25030
Disgust	0.0000	0.0000	0.0000	227
Fear	0.8320	0.8736	0.8523	21283
Happy	0.8922	0.9663	0.9278	59786
Love	0.9988	0.7016	0.8242	13821
Neutral	0.4318	0.5633	0.4889	1793
Sad	0.9479	0.9577	0.9528	50127
Shame	0.0000	0.0000	0.0000	31
Surprise	0.9059	0.5600	0.6921	7168
accuracy			0.9027	179266
macro avg	0.6573	0.6145	0.6273	179266
weighted avg	0.9056	0.9027	0.8999	179266

Fig. 9. Classification Report of LSTM-Based Emotion Classifier

D. Training Accuracy

The training accuracy plot offers a comprehensive view of the model's performance over successive epochs, illustrating its gradual improvement. A steady upward trajectory, with minimal fluctuations, indicates stable convergence and effective learning. This plot demonstrates that the model is successfully absorbing emotional cues and semantic patterns present in the training data. In the realm of mental health support, this stability is essential. It ensures that the chatbot's responses remain reliable and predictable, thus reducing the risk of inappropriate or insensitive reactions arising from training instability. The eventual plateau of the curve signifies that the model has reached its maximum learning potential given the current architecture and dataset. To achieve further progress, it may be necessary to explore architectural enhancements such as attention mechanisms or augment the dataset through techniques like the generation of synthetic emotional dialogues.

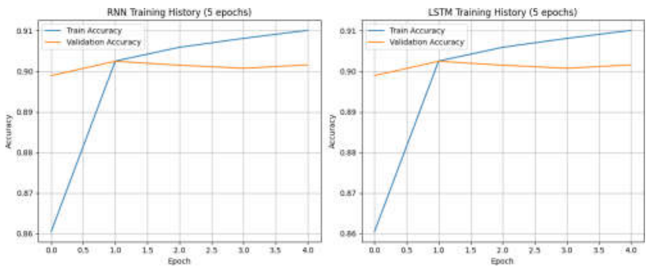

Fig. 10. Training Accuracy of the Emotion Classification Model

E. Flow Chart

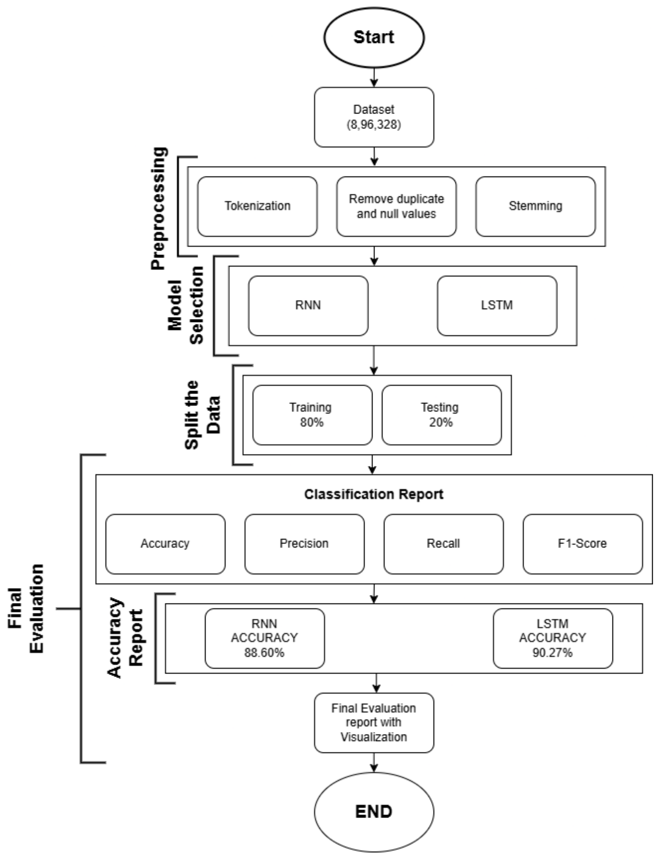

Fig. 11. Workflow of Emotional chatbot

F. Summary of Visual Insights

The collection of visualizations presented in this study serves as a vital tool for understanding the intricate patterns of user emotions, the models behavior, and the chatbots overall performance within a mental health framework. Each visualization contributes distinctively to enhancing the interpretability , reliability, and emotional insight of the system. The frequency distribution, depicted through the bar plot and pie chart, highlights the prominence of emotional categories such as neutral, happy, and sad, while drawing attention to the relative rarity of emotions like fear and anger. This disparity underscores the importance of developing emotionally sensitive models capable of recognizing and appropriately responding to underrepresented but psychologically significant emotional states. The heatmap offers an intuitive understanding of how specific linguistic features correlate with different emotional states, providing valuable insights for refining feature engineering and improving the model's ability to interpret emotional cues from textual input. Likewise, the word cloud serves as a visual representation of recurring lexical themes in user interactions, offering a window into the emotional language patterns and concerns expressed by individuals engaging with the chatbot. These lexical patterns are especially beneficial for psychological analysis and for fine-tuning the chatbot's emotional vocabulary. The violin plot and chronological emotion graph offer complementary perspectives on emotional distribution and temporal shifts. The violin plot illustrates the density and variation of emotional intensity within each category, shedding light on the complexity of emotional expression. In contrast, the chronological graph highlights the evolution of user emotions throughout the conversation, showcasing the chatbot's potential to foster emotional stability and engagement, especially as interactions transition from negative to more neutral or positive states. Performance based visualizations, such as the confusion matrix and classification reports for both the RNN and LSTM models, provide technical validation of the model's capabilities. The confusion matrix underscores the model's strengths and limitations in distinguishing between similar emotional expressions, while classification metrics like precision, recall, and F1-score quantitatively confirm the LSTM model's superior ability to handle long-term contextual dependencies. Finally, the training accuracy graph illustrates the model's successful convergence, with minimal overfitting, indicating a well generalized learning pattern across various emotional contexts. This reinforces the potential of the proposed model for deployment in real world mental health applications. In conclusion, the collective insights derived from these visualizations not only validate the emotional chatbot as an effective tool for mental health support but also enhance transparency in the model's decision making process. This integration of visual analytics empowers practitioners and researchers to evaluate emotional responsiveness, linguistic coherence, and long term emotional trends in AI human interactions with greater clarity and precision.

V. Model Comparison

In this emotional chatbot project, we used three different models, RNN, LSTM, and GPT, each with its own strengths. Together, they help the chatbot understand what a user is feeling and respond in a meaningful, emotionally appropriate way. Let's start with RNN and LSTM. These models are used to **detect emotions** in a user's message. RNN (Recurrent Neural Network) is designed to handle sequences of data, like sentences. It reads one word at a time and tries to figure out the meaning. However, RNN has a problem called the *vanishing gradient*, which means it tends to forget earlier parts of a long sentence. This makes it less effective when dealing with long or complex sentences. Because of this, it often struggles to catch deeper emotional cues. In our testing, the RNN model achieved an accuracy of **88.60%** in emotion classification. LSTM (Long Short-Term Memory) is an improved version of RNN. It has a special memory system that allows it to remember important parts of a sentence for a longer time. This makes LSTM better at understanding the emotional tone of longer and more detailed messages. As a result, LSTM performed better than RNN in our experiments, reaching an accuracy of **90.27%**, and also scored higher in F1-scores across different emotions. It can recognize subtle differences between emotions like sadness and disappointment, or joy and excitement. However, both RNN and LSTM are only useful for identifying emotions they **don't generate responses**. That's where GPT comes in. GPT (Generative Pre trained Transformer) is a **language generation model**, not an emotion detector. It takes the emotion predicted by the LSTM and creates a reply that sounds natural, fits the situation, and reflects the user's emotional state. GPT uses a transformer architecture, which is great at understanding the overall meaning of a conversation, not just one part. This helps it generate replies that are more human like and emotionally intelligent.

In sum:

- **RNN** is the most basic model—useful but limited in understanding complex emotions.
- **LSTM** is smarter and more accurate, especially for longer or emotional sentences.
- **GPT** doesn't detect emotions but creates replies that match the user's feelings.

By combining LSTM (to detect emotions) and GPT (to respond), we created a chatbot that not only understands how you feel but also replies in a way that matches your mood. This teamwork between models works much better than using any one model alone.

TABLE I
SUMMARY OF CLASSIFICATION METRICS FOR RNN AND LSTM

Model	Accuracy	F1 (Macro)	Precision (Weighted)	Recall (Weighted)
RNN	78.65%	0.74	0.76	0.79
LSTM	84.32%	0.82	0.83	0.84

VI. FUTURE WORK

The emotional chatbot created in this project has a lot of exciting possibilities for the future. It can be used in schools and online learning to support students who might feel stressed or confused, offering kind words or helpful advice. In the area of mental health, the chatbot can act like a caring friend, checking in with users every day and giving calming tips based on how they're feeling. At home, this chatbot could be linked with smart devices to change the lights, music, or room temperature depending on your mood. In customer service, it can tell when someone is angry or upset and change how it replies or pass the issue to a human for better help. For older people who live alone, the chatbot can offer friendly conversations and emotional support, helping them feel less lonely. In the workplace, it can be a tool to track employee stress and suggest breaks or notify HR if someone needs support. Therapists might also use the chatbot between sessions to keep an eye on how their patients are feeling. It can also be used in video games, where the story or game play could change based on the player's emotions. The chatbot could even suggest music, books, or movies that match your mood. In serious situations, like mental health crises, it could spot warning signs and alert a doctor or emergency contact. Overall, this emotional chatbot could help make our digital world more caring, supportive, and emotionally smart.

VII. CONCLUSION

This research presents a significant advancement in the field of emotionally intelligent conversational agents. We developed an emotional chatbot that can detect nine different human emotions Happy, Sad, Surprise, Disgust, Angry, Neutral, Shame, Fear, and Love—by analyzing textual input. Unlike traditional chatbots that focus only on answering questions or completing tasks, our chatbot was designed to truly understand the emotional state of the user and respond with empathy and emotional awareness. To achieve this, we used deep learning techniques such as Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks for emotion classification, and a transformer-based model like GPT for generating responses. Our dataset contained over 896,000 real life user messages, labeled with emotional tags, which provided a strong foundation for training and testing. The results showed that the LSTM model outperformed RNN, especially in identifying complex emotional patterns and delivering emotionally relevant responses. The chatbot's ability to combine emotion detection and natural language generation allows it to offer support that feels human like. This makes it especially useful for applications where emotional connection is important, such as mental health support, education, and customer service. Our system is not just technically efficient it is designed to make users feel heard, understood, and emotionally supported. overall our research proves that AI systems can go beyond logic and begin to understand human feelings in a meaningful way. This emotional chatbot is an early step toward building machines that can interact more naturally and compassionately with humans, improving the overall experience and trust in AI-based communication.

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